FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Advanced)-2020

PAPER – 2

TEST DATE: 15-12-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A, C, D

Sol. Amplitude = $\frac{13}{2}$ m, $\omega = \frac{2\pi}{T} = \frac{\pi}{4}\,s^{-1}$

$$\vec{r} = \frac{13}{2}(1-\cos\omega t)\frac{12}{13}\hat{i} + \left[5 - \frac{13}{2}(1-\cos\omega t)\frac{5}{13}\right]\hat{j}$$

$$\vec{r} = 12\sin^2\left(\frac{\pi t}{8}\right)\hat{i} + 5\cos^2\left(\frac{\pi t}{8}\right)\hat{j}$$

$$\therefore\ \vec{v} = \left[\frac{3\pi}{2}\sin\left(\frac{\pi t}{4}\right)\hat{i} - \frac{5\pi}{8}\sin\left(\frac{\pi t}{4}\right)\hat{j}\right] m/s$$

2. A, B, C

Sol. By using basic concept of fluid mechanics.

3. A, B, C

Sol. Let acceleration of ball relative to the cube is $\vec{a}_{b,c} = a'_x\hat{i} + a'_y\hat{j} + a'_z\hat{k}$

Now, $a'_x = \left(\frac{\rho - \frac{\rho}{3}}{\frac{\rho}{3}}\right) a_x = 2a_x = 2 \times \frac{5}{\sqrt{2}} = 5\sqrt{2}\ m/s^2$

$a'_y = \left(\frac{\rho - \frac{\rho}{3}}{\frac{\rho}{3}}\right)(g + a_y) = 2(g + a_y) = 2\left(10 - \frac{15}{2}\right) = 5\ m/s^2$

$a'_z = \left(\frac{\rho - \frac{\rho}{3}}{\frac{\rho}{3}}\right) a_z = 2a_z = 2 \times \frac{5}{2} = 5\ m/s^2$

$\therefore$ Acceleration of ball relative to the cube is $\vec{a}_{b,c} = (5\sqrt{2}\hat{i} + 5\hat{j} + 5\hat{k})\ m/s^2$

4. A, C, D

Sol. $T = 0.2 = 2\pi\sqrt{\frac{m}{k}}$

$\Rightarrow m = 0.2$ kg

At t = 0, particle is at extreme position and moving towards +ve x-axis

$K_{max} = \frac{1}{2}m(v_{max})^2 = 4.0\ J$

5. A, B, C

Sol. $\frac{m_0 v^2}{r_n} = \frac{1}{4\pi\varepsilon_0}\frac{ze^2}{r_n^2}$ (Where $m_0 = 50m$) …(i)

$m_0 v r_n = \frac{n}{2}\frac{h}{\pi}$ …(ii)

$r_n = \frac{n^2 h^2 \varepsilon_0}{200\pi m e^2}$

$E_n = \frac{-13.6z^2}{n^2} \times 50\ eV$

$E_4 = -680$ eV

$E_1 = -10880$ eV

$\Delta E = E_4 - E_1 = 10{,}200$ eV

$\lambda = \frac{1240}{\Delta E} = 1.21\ \text{Å}$

6. A, B, D

Sol. $F_x = \frac{IA\cos\theta\sin\theta}{C}(1 - r) = 1.152 \times 10^{-6} N$

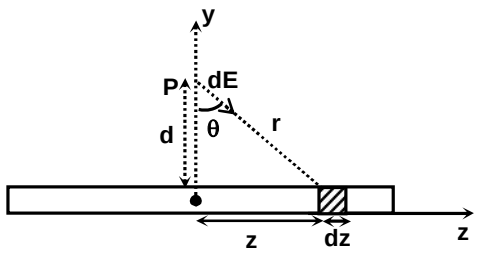
$F_y = \frac{IA\cos^2\theta}{C}(1 + r) = 2.016 \times 10^{-6} N$

$p_r = \frac{I\cos^2\theta}{C}(1 + r) = 1.008 \times 10^{-6}\ N/m^2$

7. A, B, C, D

Sol. Gravitational field due to a long wire,

$$dE = \frac{2G\lambda}{r} = \frac{2G\sigma dz}{r}$$



$$E_P = \int dE.\cos\theta = \int \frac{2G\sigma dz}{r}\frac{d}{r} = 2G\sigma d\int_{-\frac{a}{2}}^{+\frac{a}{2}}\left(\frac{dz}{d^2+z^2}\right)$$

$$E_P = 4G\sigma\tan^{-1}\left(\frac{a}{2d}\right)$$

8. A, B, C, D

Sol. For the first interference maximum

$$2d\sin\theta = \frac{\lambda}{2}$$

$$\Rightarrow \sin\theta = \frac{\lambda}{4d} = \frac{10^{-3}}{4}$$

$$\frac{y}{D} = \frac{10^{-3}}{4}$$

$$y = \frac{1}{4}\text{mm}$$

Fringe width, $\omega = \frac{\lambda D}{2d} = 0.5\,\text{mm}$

$$2d\sin\theta = \frac{\lambda}{4}$$

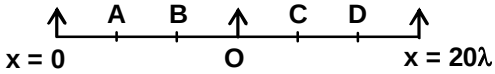
$$\Rightarrow \sin\theta = \frac{\lambda}{8d}$$

$$\therefore y = \frac{\lambda D}{8d} = \frac{1}{8}\text{mm}$$

9. B

10. D

Sol. **(for Q. 9-10):** Path difference = integral multiple of λ in all cases i.e. interference is constructive.



$$I_0 = \frac{36}{100}kA^2,\ I_A = \frac{9}{16}kA^2,\ I_B = \frac{49}{144}kA^2,\ I_C = \frac{4}{9}kA^2 \text{ and } I_D = \frac{81}{64}kA^2$$

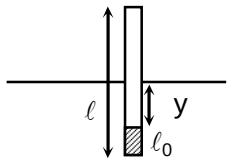
For second case

$$I_A = \frac{81}{64}kA^2,\ I_B = \frac{4}{9}kA^2,\ I_C = \frac{49}{144}kA^2 \text{ and } I_D = \frac{9}{16}kA^2$$

11. A

12. D

Sol. **(for Q. 11-12)**



$\ell_0 A \times 2.7 + \ell A \times 0.8 = (\ell_0 + y)A$

$\ell_0(2.7 - 1) + 0.8\ \ell = y$

$y = 1.7\ \ell_0 + 0.8\ell$

$y = 1.7 + 0.8\ \ell$

at t = 5 minute; ℓ = 9.5 cm y = 9.3

at t = 10 minute; ℓ = 9 cm y = 8.9

at t = 15 minute; ℓ = 8.5 cm y = 8.5

at t = 20 minute; ℓ = 8 cm y = 8.1 which is not possible $\Rightarrow$ y = 8.5

$$T = 2\pi\sqrt{\frac{m}{A\rho g}}$$

$m_0 = k(2.7 + 8) = 10.7\ k$
$m_5 = k(2.7 + 7.6) = 10.3\ k$
$m_{10} = k(2.7 + 7.2) = 9.9\ k$
$m_{15} = k(2.7 + 6.8) = 9.5\ k$
$m_{20} = k(2.7 + 6.8) = 9.5\ k$

$T_0 \propto \sqrt{10.7}$

$T_5 \propto \sqrt{10.3} \Rightarrow \sqrt{\frac{103}{107}}T_0$

$T_{10} = \sqrt{\frac{99}{107}}T_0$

$T_{15} = \sqrt{\frac{95}{107}}T_0$

$T_{20} = \sqrt{\frac{95}{107}}T_0$

SECTION – C

13. 00005.55

Sol. $\beta = 10 \log \frac{I}{I_0}$

$$3 = \log\frac{I}{10^{-12}}$$

$\Rightarrow I = 10^{-9}\ W/m^2$

$$\text{Now, } I = \frac{(\Delta P_0)^2 v}{2B} = \frac{(BAK)^2 v}{2B} = \frac{B\omega^2 A^2}{2v} = \frac{BA^2 4\pi^2 f^2}{2v}$$

$$A = \sqrt{\frac{Iv}{B2\pi^2 f^2}} = \sqrt{\frac{I}{\rho v 2\pi^2 f^2}} = 5.55\ \text{Å}$$

14. 00000.20

Sol. Let mass of P, Q and S are m_1, m_2 and m_3 respectively

$$\frac{Gm_1}{R^2} = \frac{G(m_1 + m_2 + m_3)}{9R^2}$$

$8m_1 = m_2 + m_3$

$8\rho_1\frac{4}{3}\pi R^3 = \frac{4}{3}\pi R^3(7\rho_2 + 19\rho_3)$

On solving

$\frac{\rho_3}{\rho_1} = 0.2$

15. 00012.00

Sol. $\lambda_A = \lambda_0 + v_s T = vT + \frac{v}{3}T = \frac{4}{3}\frac{v}{f}$

$\lambda_B = \frac{\frac{4}{3}\frac{v}{f}}{\frac{1}{4}} = \frac{16}{3}\frac{v}{f}$

16. 00489.51

Sol. $\Delta W_{ex}\left(\int_0^a \rho a^2 x g dx\right) + \rho a^3 g(h-a) = \rho a^3 g\left(h - \frac{a}{2}\right) = 00489.51$

17. 00001.12

Sol. $\vec{F} = \frac{d\vec{P}}{dt} = \frac{\Delta\vec{P}}{dt}$

$\left|\vec{F}\right| = \sqrt{2}\cdot u\frac{dm}{dt} = \sqrt{2}\times 5\sqrt{2}\times 0.112 = 1.12$ newton

18. 00097.50

Sol. $\sin i = \frac{4}{3}\sin r$

$\therefore$ r = 37°

Now, $2\mu t \sec r = \lambda/2$

$\Rightarrow t = \frac{\lambda}{4\mu \sec r} = 97.50$ nm

Chemistry

PART – II

SECTION – A

19. A, B, C, D

Sol. $\left[Co(NH_3)_6\right]^{3+}$ - Octahedral and diamagnetic $\left[Fe(NH_3)_6\right]^{2+}$ - Octahedral and paramagnetic.

$\left[Pt(CN)_4\right]^{2-}$ - Square planar, $\left[CuCl_4\right]^{2-}$ - Tetrahedral

$\left[Ni(NH_3)_6\right]^{2+}$ - Primary valency = 2

$\left[Cr(NH_3)_6\right]^{3+}$ - Primary valency = 3

20. A, B, C

Sol. MnO_4^- is coloured due to charge transfer phenomena.

21. A, B, D

Sol. $OCl^- > ClO_2^- > ClO_3^- > ClO_4^-$ (Cl – O bond distance)

$HClO_4 > HBrO_4 > HIO_4$ (Acidity)

H – F > H – Cl > H – Br > H – I (Bond energy)

22. A, B, C

Sol. $2Cu^{2+} + K_4\left[Fe(CN)_6\right] \longrightarrow \underset{\text{Chocolate brown ppt.}}{Cu_2\left[Fe(CN)_6\right]\downarrow}$

$3Cu^{2+} + 2K_3\left[Fe(CN)_6\right] \longrightarrow \underset{\text{Green ppt.}}{Cu_3\left[Fe(CN)_6\right]_2\downarrow}$

$Hg_2Cl_2 + 2NH_4OH \longrightarrow \underbrace{Hg + Hg(NH_2)Cl}_{\text{Black residue}} + NH_4Cl + 2H_2O$

23. A, B, C, D

Sol. (i) White phosphorous is more reactive than red phosphorous.

(ii) P_2H_4 is less stable than N_2H_4

(iii) Bond angle in PH_4^+ is higher than that in PH_3

(iv) In the solid state PBr_5 exists as $\left[PBr_4\right]^+Br^-$

24. A, B, C

Sol. Sulphide ores are concentrated by froth floatation process.

25. A, B, C

Sol. $N_3P_3Cl_6$, C_3N_{12} and N_4S_4 have cyclic structure.

26. A, B, C

Sol. Azeotropes do not obey Raoult's law.

27. B

Sol. $\left[Co(NH_3)_6\right]^{3+}$ is an inner orbital complex ion. Hybridisation of central atom is d^2sp^3.

28. A

Sol. $\left[COF_6\right]^{3-}$ is an outer orbital complex ion. Hybridization of the central atom is sp^3d^2.

29. A

Sol. NO_2 is a brown coloured gas, acidic oxide and paramagnetic.

30. C

Sol. NO is a colourless gas, neutral oxide and paramagnetic.

SECTION – C

31. 00045.15

Sol. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

$a \qquad 0$

$a-x \qquad 2x$

$\frac{2x}{a-x} = 2, \; x = \frac{a}{2}$

$t = \frac{2.303}{92.12\times10^{-2}} \log \frac{a}{a-x}$

$t = \frac{2.303}{92.12\times10^{-2}} \log 2$

$t = \frac{2.303\times0.3010}{92.12\times10^{-2}}$

$t = 0.7525 \text{ min} = 45.15 \text{ sec}.$

32. 00023.84

Sol. $3CuO(s) + 2NH_3(g) \longrightarrow 3Cu(s) + N_2(g) + 3H_2O(\ell)$

$\Delta rH^o = \sum \Delta H_f^o(\text{Products}) - \sum \Delta H_f^o(\text{Reactants})$

= [3 × (- 285)] – [3 × (- 155) + 2 × (- 46)]

= - 298 kJ

Heat liberated $= \frac{298\times2.72}{2\times17} = 23.84$ kJ

33. 00031.25

Sol. $\Delta T = iK_b \; m$

$5 = 1.8\times0.52\times\frac{1000}{100}\times\frac{W}{58.5}$

$W = \frac{5\times58.5}{1.8\times0.52}$

$W = 31.25 \text{ g}$

34. 00033.25

Sol. $\frac{3}{M+8} = \frac{5}{M+35.5}$

$M = 33.25$ (M = EW of metal)

35. 00044.80

Sol. $P_{O_2} = P_{oxide}$

$$\frac{P_1M_1}{RT_1} = \frac{P_2M_2}{RT_2}$$

$$\frac{4.2 \times 32}{300} = \frac{3.2 \times M_2}{320}$$

$M_2 = 44.80$

36. 00002.60

Sol. $\lambda^o_{AgCl} = \lambda^o_{Ag^+} + \lambda^o_{Cl^-}$

$= 62.3 + 67.7 = 130$

$$\lambda^o_{AgCl} = K \times \frac{1000}{M}$$

$$130 = 3.38 \times 10^{-6} \times \frac{1000}{M}$$

$$M = \frac{3.38 \times 10^{-6} \times 1000}{130}$$

$$M = 2.6 \times 10^{-5} \text{ M}$$

$$\therefore y = 2.60$$

Mathematics

PART – III

SECTION – A

37. A, B, C

Sol. Let $\alpha, \beta, \gamma, \delta$ be the roots $\Rightarrow \alpha + \beta + \gamma + \delta = -p$, $p < 0$

$\Sigma\alpha\beta = q$, $q > 0$, $\Sigma\alpha\beta\gamma = -r$, $r < 0$, $\alpha\beta\gamma\delta = 5$, AM $\geq$ G.M.

$$\Rightarrow \frac{\Sigma\alpha}{4} \geq (\alpha\beta\gamma\delta)^{\frac{1}{4}} = 5^{\frac{1}{4}},\ -p \geq 4\cdot 5^{\frac{1}{4}}$$

Also, $\frac{\Sigma\alpha\beta\gamma}{4} \geq (\alpha\beta\gamma\delta)^{\frac{3}{4}} = 5^{\frac{3}{4}} \Rightarrow -r \geq 4\cdot 5^{\frac{3}{4}} \Rightarrow pr \geq 16.5 = 80$

38. C

Sol. $(a^2 + b^2)^3 + 32(a^3 + b^3)(a + b) \geq 32(a^3 + b^3)ab$

hold true if $ab < 0$

Let $a.b \geq 0 \Rightarrow a, b \geq 0$

Using A.M $\geq$ G.M

$(a^2 + b^2)^3 + 32(a^3 + b^3)(a + b) \geq 2((a^2 + b^2)^3\ 32(a^3 + b^3)(a + b))^{1/2}$

$$= 8\sqrt{2\left(a^2 - ab + b^2 + ab\right)^3\left(a^3 + b^3\right)(a+b)} \geq 8\sqrt{2\left(2\sqrt{\left(a^2 - ab + b^2\right)(ab)}\right)^3\left(a^3 + b^3\right)(a+b)}$$

$$= 32\sqrt{\left(a^3 + b^3\right)^2\sqrt{\left(a^2 - ab + b^2\right)a^3b^3}} \geq 32\sqrt{\left(a^3 + b^3\right)^2\sqrt{aba^3b^3}} = 32(a^3 + b^3)ab$$

39. C

Sol. L.H.S. $= \sum_{r=1}^{n}(-1)^{r-1}\left(\frac{1}{r} + \frac{1}{(r+1)^2}\right){}^nC_r$, $(1-x)^n = {}^nC_0 - {}^nC_1x + {}^nC_2x^2 - {}^nC_3x^3 + \ldots + (-1)^n\,{}^nC_nx^n$

$$\int_0^1 \frac{1-(1-x)^n}{x}dx = \int_0^1\left({}^nC_1 - {}^nC_2x + {}^nC_3x^2 + \ldots + (-1)^{n-1}\,{}^nC_nx^{n-1}\right)dx$$

$$1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n} = \sum_{r=1}^{n}(-1)^{r-1}\left(\frac{{}^nC_r}{r}\right)$$

Similarly, $\int_0^1 \frac{1-(1-x)^{n+1}}{x(n+1)}dx = \int_0^1\left(\frac{{}^{n+1}C_0}{1} - \frac{{}^{n+1}C_1}{2}x + \frac{{}^{n+1}C_2}{3}x^2 + \ldots + (-1)^{n+1}\frac{{}^{n+1}C_n}{n+1}x^{n+1}\right)dx$

$$\Rightarrow \left(\frac{1}{n+1}\right)\left(1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n+1}\right) = 1 - \frac{{}^nC_1}{2^2} + \frac{{}^nC_2}{3^2} + \ldots + (-1)^n\frac{{}^nC_n}{(n+1)^2} = \sum_{r=1}^{n}(-1)^{r-1}\frac{{}^nC_r}{(r+1)^2}$$

$$\therefore\ y = \left(\frac{n}{n+1}\right)\left(1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}\right),\ \frac{n+1}{n} \leq \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n} \leq \frac{n+1}{1}$$

40. A, B

Sol. Intersection of branch of hyperbola with the circle.

41. B, D

Sol. Equation of plane containing the line with direction cosines (l, m, n) and the line $\frac{x}{1} = \frac{y}{\tan\alpha} = \frac{z-c}{0}$

is $\begin{vmatrix} x & y & z-c \\ 1 & \tan\alpha & 0 \\ l & m & n \end{vmatrix} = 0$

$\Rightarrow (n\tan\alpha)x - ny + (m - l\tan\alpha)(z - c) = 0$ ..... (1)

Equation of plane containing the line with direction cosines (l, m, n) and the line

$\frac{x}{1} = \frac{y}{-\tan\alpha} = \frac{z+c}{0}$ is $\begin{vmatrix} x & y & z+c \\ 1 & -\tan\alpha & 0 \\ l & m & n \end{vmatrix} = 0$

$\Rightarrow$ $(-n \tan \alpha)x - ny + (m + l \tan \alpha)(z + c) = 0$ ..... (2)

$\therefore$ Plane through (1) and (2) is

$(n \tan \alpha\, x - ny + (m - l \tan \alpha)(z - c)) + \lambda(n \tan \alpha x + ny - (m + l \tan \alpha)(t + c)) = 0$ is parallel to z-axis

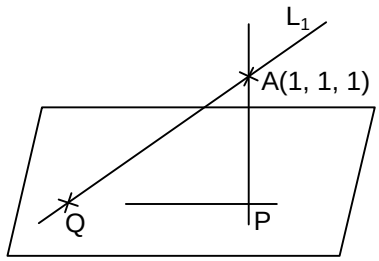
Hence, $(m - l \tan \alpha) - \lambda(m + l \tan \alpha) = 0 \Rightarrow \lambda = \left(\frac{m - l\tan\alpha}{m + l\tan\alpha}\right)$

$\therefore$ Shortest distance is = $\left|\frac{c\left(m^2 \cot\alpha - l^2 \tan\alpha\right)}{n\sqrt{m^2 + l^2}}\right|$.

42. B, D

Sol. Any point on line $L_1 : \frac{x-1}{2} = \frac{y-1}{1} = \frac{z-1}{1}$ is

(2r + 1, r + 1, r + 1)



Put in equation of plane $x + y + z = 1 \Rightarrow r = -\frac{1}{2}$

$\therefore$ $Q = \left(0, \frac{1}{2}, \frac{1}{2}\right)$. For point P equation of AP

$\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1} = t$ $\therefore$ Any point on line (t + 1, t + 1, t + 1) lie to plane P

$\Rightarrow$ $t = -\frac{2}{3}$ $\Rightarrow$ Point P = $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

$\therefore$ Equation of QP $\frac{x - \frac{1}{3}}{\frac{1}{3}} = \frac{y - \frac{1}{3}}{-\frac{1}{6}} = \frac{z - \frac{1}{3}}{-\frac{1}{6}}$

$\Rightarrow$ $\frac{x - \frac{1}{3}}{2} = \frac{y - \frac{1}{3}}{-1} = \frac{z - \frac{1}{3}}{-1}$

43. C, D

Sol. $2^{n-1}(2^n + 1)$

Let $a_n \Rightarrow$ the number of arrangements at n places such that **'a'** occurs even number of times and $b_n \Rightarrow$ the number of arrangements at n places such that **'a'** occurs odd number of times

$\Rightarrow a_n = b_{n-1} + 3a_{n-1} = 4^{n-1} + 2a_{n-1}$

$\Rightarrow a_n - 2a_{n-1} = 4^{n-1}$

Let $a_n = 2^n t_n \Rightarrow 2^n(t_n - t_{n-1}) = 2^{2n-2} \Rightarrow t_n - t_{n-1} = 2^{n-2}$

$\Rightarrow$
$t_2 - t_1 = 2^0$
$t_3 - t_2 = 2^1$
$t_4 - t_3 = 2^2$
$\vdots$
$t_n - t_{n-1} = 2^{n-2}$

where $t_1 = \frac{3}{2}$

$t_n - t_1 = \left(2^{n-1} - 1\right)$

$\therefore$ $a_n = 2^{n-1}(2^n + 1)$

44. B, D

Sol. $A = \begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix} = 0$, $R_3 \to R_3 - \alpha R_1 - R_2$

$A = \begin{vmatrix} a & b & 0 \\ b & c & 0 \\ a\alpha+b & b\alpha+c & a\alpha^2+2b\alpha+c \end{vmatrix} = 0 \Rightarrow (a\alpha^2 + 2b\alpha + c)(ac - b^2) = 0$

$\Rightarrow$ Either $b^2 = ac$ (or) $a\alpha^2 + 2b\alpha + c = 0$

45. B

46. C

Sol. (for Q. 45.-46.)

$0 < x_i < 1 \Rightarrow x_i \Rightarrow \frac{1}{30}, \frac{7}{30}, \frac{11}{30}, \frac{13}{30}, \frac{17}{30}, \frac{19}{30}, \frac{23}{30}, \frac{29}{30} \Rightarrow \Sigma x_i = 4$

for $1 < x_i < 2$ $\quad x_i : \frac{1}{30}+1, \frac{7}{30}+1, \frac{11}{30}+1, ..., \frac{29}{30}+1 \Rightarrow \Sigma x_i = 12$

Similarly for $4 < x_i < 5$ $\frac{1}{30}+4, \frac{7}{30}+4, \frac{11}{30}+4, ..., \frac{29}{30}+4$

Hence $\sum_{i=1} x_i = 4 + 12 + 20 + ...$ upto n term

$= \frac{n}{2}(8 + (n-1)\,8) = \frac{n}{2} \cdot 8n = 4n^2$

47. C

48. D

Sol. $|\vec{r} - \vec{b}| + |\vec{r} - \vec{c}| = 4$ represent, an ellipsoid in 3D space with a = 2 and $|\vec{b} - \vec{c}| = 2ae$

$\Rightarrow b = \frac{1}{\sqrt{2}}$, $\vec{r} \cdot \vec{a} = 5$ is plane passing through $\vec{b}$, $\vec{c}$

Hence, area B area of ellipse = $\pi ab = \pi \cdot \frac{2}{\sqrt{2}} = \sqrt{2}\,\pi$ square unit

SECTION – C

49. 00589.00

Sol. Let $n^2 = 2^{62}.3^{38}$

Let F is a factor or $n^2 \Rightarrow 2^\alpha\, 3^\beta$, $0 \le \alpha < 62$; $0 \le \beta \le 38$

$\Rightarrow$ n(F) = $63 \times 39 = 2457$

Let a, b are two factor, of n^2 such that $a.b = n^2$ when $a, b \ne n$

if $a < n$ then $b > n$ will exist in pairs

$\therefore$ number of factors less than n = 1228

Similarly if x and y are factors of n such that

x.y = n and x and y are also factors of n^2

$\Rightarrow$ number of such factors = $(32 \times 20 - 1) = 639$

$\therefore$ req. number of factors = $1228 - 639 = 589$

50. 00000.72

Sol. $n(E) = 4 \times 9 \times 10^3 = 36000$

n(S) = 50,000

$\Rightarrow$ P(E) = 0.72

51. 00002.50

Sol. Let $y = x + \frac{1.3}{2!}x^2 + \frac{1.3.5}{3!}x^3 + \ldots\ldots\ldots\ldots$

$$\frac{dy}{dx} = 1 + \frac{1.3}{1!}x^1 + \frac{1.3.5}{2!}x^2 + \ldots\ldots\ldots\ldots$$

$$x\frac{dy}{dx} = x + \frac{1.3}{1!}x^2 + \frac{1.3.5}{2!}x^3 + \ldots\ldots\ldots\ldots$$

and $y - \frac{dy}{dx} = -1 + \left(1 - \frac{3}{1!}\right)x + \frac{1.3}{2!}(1-5)x^2 \ldots\ldots\ldots$

$$y - \frac{dy}{dx} = -1 - 2\left(x\frac{dy}{dx}\right)$$

$$\int\frac{dy}{1+y} + \int\frac{dx}{2x-1} = 0$$

$\sqrt{1-2x}(1+y) = 1$ put x = 2/5 hence $y = \sqrt{5} - 1$

52. 00001.00

Sol. R.H.S. $\geq 1 \Rightarrow$ L.H.S. ≥ 1

$\Rightarrow y > |x| \geq 0 \Rightarrow y^2 + x^2 - 2|x|y \geq x^2 + y^2 + 2\sqrt{x^2+y^2-1}$

$-|x|y \geq \sqrt{x^2+y^2-1}$, L.H.S. ≤ 0 and R.H.S. ≥ 0.

Hence, L.H.S. = R.H.S. = 0

$\Rightarrow |x|y = 0$ and $x^2 + y^2 - 1 = 0$, $y > 0$

$\Rightarrow$ x = 0 and $y^2 = 1 \Rightarrow y = \pm 1$. Since $y > 0 \Rightarrow y = 1$

Hence, any one order pair of (x, y) = (0, 1)

53. 00000.25

Sol. $a(n) = (x^n + y^n)(x^2 + y^2) = x^{n+2} + y^{n+2} + x^2y^n + y^2x^n$

$a(n) = a(n + 2) = x^2y^2(x^{n-2} + y^{n-2})$, $\frac{a(n) - a(n+2)}{a(n-2)} \leq x^2y^2 = x^2(1-x^2)$

$= x^2 - x^4 = \frac{1}{4} - \left(\frac{1}{2} - x^2\right)^2 \leq \frac{1}{4} \Rightarrow \frac{a(n) - a(n+2)}{a(n-2)} \leq \frac{1}{4}$

54. 00001.25

Sol. Line of intersection of two planes and x-axis must be coplanar i.e., (x, 0, 0) must satisfy both the planes

$\Rightarrow 5x = d_1$ and $4x = d_2$, $\frac{d_1}{d_2} = \frac{5}{4} = 1.25$